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Is it time to go for no-car zone policies? Braess Paradox Detection

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ABSTRACT

No-car zone is becoming a trend in many parts of the world, mainly driven by environmental motives. Recent data has shown that both vehicle kilometers traveled and car ownership are on the decline, especially in developed countries. This has been coupled with the rise of new urban mobility concepts such as smart mobility, car sharing, and green modes. Can we push for these policies without compromising traffic circulation? The Braess Paradox (BP) suggests that road closure can, in fact, improve traffic congestion; dubbed as the Braess Paradox Detection Problem (BPDP). This study aims to leverage the BP to boost the idea of the no-car zone. To this end, we formulate the BPDP as a bilevel problem and develop a heuristic methodology (a surrogate-based algorithm) to identify roads for closure in the heart of the cities to be reclaimed for green space, pedestrian plazas, and so on. Furthermore, our proposed formulation considers travel demand elasticity resulting from road-closure schemes while solving for the BPDP. We test the methodology using a real dataset, the road network of Winnipeg. The results indicate strong evidence and justification supporting the idea of no-car zone even in the heart of highly congested cities.

1. Introduction

A large part of London's flagship Oxford Street could go car-free by 2018 and become a pedestrian plaza (Garfield, 2017). Other European cities such as Madrid, Paris, and Oslo have also heralded similar schemes. Recently, Germany's highest administrative court ruled that in an effort to improve urban air quality, cities may ban cars from some streets (Eddy, 2018). While their motive is largely rooted in environmental concerns, there exist yet other strong justifications advocating the following:

(i) In the wake of smart cities and communication technologies, the definition of accessibility needs to be revisited. Motorized traffic in general and private vehicles, in particular, are not the only means that provide "accessibility"; smartphone-driven ideas such as smart bike-sharing systems (Liu et al., 2016), and ride-sharing services have become reliable sources of transportation (Fang, 2015). Recent data indicates that vehicle kilometer traveled as an index for the number of cars on the roads is declining (Fang, 2015).
(ii) Closing some roads can, in fact, reduce traffic congestion, in a counterintuitive phenomenon known as the Braess Paradox (BP) (Braess, 1968; Braess et al., 2005). In other words, the idea of BP can provide advocates of banning cars with extra ammunition,

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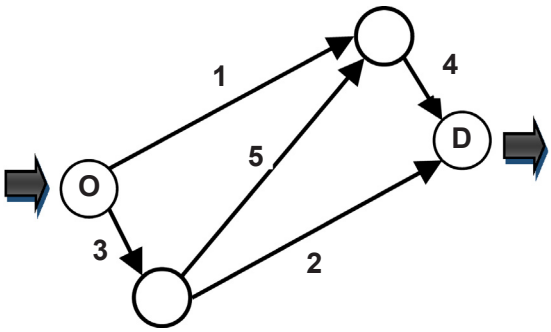

Fig. 1. The famous Braess example.

not from the altruistic point of view, but from a genuine transport-engineering angle, that is, the idea can, in fact, reduce traffic congestion.

(iii) Roads restrictions per se is not a new concept and it has been around in many forms and shapes. Transit priority and bicycle lanes are of such schemes, in which the idea is to take some space from private cars to be exclusively used by public transport (metro, buses etc.) or bicycles. The same can be said about traffic detour planning (Bagloee et al., 2014) and congestion pricing (charging drivers using certain roads) as well as parking restrictions (Bagloee et al., 2012).

Is the BP limited only to transport systems? In addition to transport, the BP phenomenon has also been observed in other fields such as computer networks (Korilis et al., 1999), water distribution pipes (Calvert and Keady, 1993), springs in mechanical engineering (Penchina and Penchina, 2003), queue flow (Inoie et al., 2006), semiconductors in electrical circuits (Sellier et al., 2011), and crowd control (Hughes, 2003).

What causes the BP? Drivers (or users of a transport system) "selfishly" choose the fastest routes (or shortest paths), resulting in a traffic pattern called "user equilibrium" (UE). A better traffic pattern (in terms of the total travel time spent by all users in the system) can be shaped if drivers cooperate with one another when choosing their routes in order to minimize the system's cost, known as system equilibrium or system optimum (SO). The BP lies in the selfish behavior of the users, who seek out the paths that minimize their own travel time even if it results in greater travel times for other users. For example, if they find the shortcut road number 5 shown in Fig. 1, they do not hesitate to drive through, which may leave other users worse off. Rapoport et al. (2009) use real driving scenarios in laboratory experiments that have shown a strong correlation between the occurrence of the BP and drivers' behavior. Therefore, the idea is to detect these unhealthy shortcuts (such as Braess-tainted road number 5) and close them in an attempt to address traffic congestion.

How big or devastating can the BP be? Koutsoupias and Papadimitriou (1999) coin the term "price of anarchy" (PA), which represents the ratio of system's cost using UE's cost divided by SO's cost. The PA for the transport network has been shown to be as high as 2.15 (Roughgarden, 2005, 2006; Roughgarden and Tardos, 2002). In other words, such a deep gap between the existing traffic pattern (UE) and an ideal pattern (SO) provides a convincing motive and justification for transport authorities and planners to explore techniques such as traffic restriction, congestion pricing, road closure, transport priority, and high-occupancy vehicle lanes aiming to improve traffic circulation (Bagloee and Sarvi, 2017; Bagloee et al., 2017a, 2016).

How prevalent is the BP in transport? Is BP just an intriguing on-paper riddle that is observed in the lab and in computer simulations, or does it really exist in the world? To answer these questions, let us consider the following:

- In Seoul, Korea, the destruction of a six-lane highway to build a public park improved travel time into and out of the city, even though traffic volume stayed roughly the same before and after the change (Easley and Kleinberg, 2010).
- In Boston, six possible road closures, including parts of Charles and Main Streets, reduced delays (Baker, 2009).
- In Stuttgart, Germany, after investments in the road network in 1969, the traffic situation did not improve until a section of a newly built road was closed to traffic again (Knödel, 2013).
- In 1990, the temporary closing of 42nd Street in New York City for Earth Day reduced the amount of congestion in the area (Kolata, 1990).
- Youn et al. (2008) analyzed the travel times in road networks of several major cities, including London, Boston, and New York. They identified several streets whose closure can partially improve the traffic conditions.
- In 2009, New York experimented with closures of Broadway at Times Square and Herald Square, which resulted in improved traffic flow and hence turning the road permanently to pedestrian plazas (Wikipedia, 2018).
- Cairns et al. (1998) identified over 70 cases in which restricting traffic or closing roads has resulted in reduced traffic congestion.
- Many other examples indicate that BP is an omnipresent and latent phenomenon in the real world. Hence road closure can offer a tangible improvement on the performance of real-world networks (Kelly, 2008; Pas and Principio, 1997; Roughgarden, 2005; Steinberg and Zangwill, 1983).
- As the size of a transport network increases (more roads and intersections), so too does the likelihood of the BP occurrence (Chotras, 2015). In other words, in the real world (large-sized networks), the BP is prevalent.

In addition to environmental concerns and traffic congestion, there is another and yet very interesting angle to the road-closure schemes. In Canada, to reduce future budgetary strain, municipalities have begun assessing options to shut down some transportation assets, including roads, due to their high maintenance expenses (Burke et al., 2014). But which roads should be closed such that their closures do not exacerbate traffic circulation? This is the research question that is addressed in this study. To this end, we formulate the Braess Paradox Detection Problem (BPDP) as a bilevel problem in which the upper level minimizes the total travel time (the system's cost) while accounting for the driver's behavior (UE traffic pattern) at the lower level. Any change to the road network may result in a change in travel time of some drivers, which, in turn, may affect the travel demand (e.g. some may choose to change their departure time or even cancel their trip). This is called elastic travel demand, which is thoroughly accounted for in the proposed formulation. The BPDP and bilevel problem are proved to be of highest computational complexity, known as an NP-hard problem (Jeroslow, 1985; Roughgarden and Tardos, 2002). To this end, as a solution algorithm, we employ a surrogate-based heuristic algorithm (Bagloee et al., 2018b; Wismans et al., 2014), which has been found to be highly efficient and effective for large-sized NP-hard problems (Bagloee et al., 2018a, 2018c). We then test the methodology on the large-sized network of Winnipeg, Canada.

The paper is organized as follows. Section 2 provides a review of the relevant studies. In Section 3, the mathematical setup of the methodology is articulated. In Section 4, we discuss details of the ML-O solution algorithm. Numerical evaluations are provided in Section 5. We conclude the paper in Section 6.

2. Literature review

The notion of BP has inspired many scholars to look at network flow in general from different angles. Bagloee et al. (2017d) and Zhao et al. (2014) have looked at the transport robustness and vulnerability from the angle of the BP. Zhao et al. (2014) show that as the congestion increases, the severity of the BP first increases and then drops. Jiang and Szeto (2016) show that the BP may occur in transit assignment modeling, that is, adding a new bus line or increasing the number of vehicles in a public transport fleet may result in greater travel time for passengers. Wang and Szeto (2017) and Nagurney (2000) have shown that BP-like phenomena can occur in noise and pollution emission. Yin and Ieda (2002) show that increasing the capacity or decreasing the variability of travel time for some roads may lead to a less-reliable network.

As noted before, the BP is related to the drivers' routing behavior which itself is an outcome of the interaction between supply (road network) and (travel) demand. In other words, any changes to either side of this equation (i.e., supply vs. demand) can result in a BP or its disappearance (Aashtiani and Poorzahedy, 2004; Pas and Principio, 1997; Wang et al., 2016; Yin and Ieda, 2002). The link between the supply and demand is governed by the roads' volume-delay functions (VDFs). In other words, the type of VDF does significantly affect the likelihood and intensity of the BP. For linear VDF, the PA has been shown to be as high as 4/3, whereas, for nonlinear functions (as is the case in transport models), it can reach to 2.15 (Roughgarden and Tardos, 2002). Hence, the BP and the PA present an ample opportunity for traffic authority to improve traffic circulation by deviating from UE and drifting toward SO for which there are two general approaches: (i) either to impose a toll on Braess-tainted roads, or (ii) simply to close them. Due to operational complexity of the former, some scholars suggest the latter (Cole et al., 2006; Fotakis et al., 2017, 2012), known as Braess Paradox Detection Problem (BPDP), which is the subject of this paper. Given the complexity of the BPDP (i.e., NP-hard), several heuristic methodologies can be found in the literature.

Some researchers have attempted to address the BPDP indirectly when designing the road network. Yang and Bell (1998) propose the idea of network reserve capacity. Burke et al. (2014) employ a sensitivity analysis approach. Bagloee et al. (2017b) develop a branch-and-bound tree to avoid the BP by tracking down the solutions over the tree. Korilis et al. (1999) have proposed some rule-of-thumb guidelines to avoid the BP when adding new capacity to a network.

Some scholars have addressed the BPDP for very special cases, such as single commodity networks (Fotakis et al., 2017; Kojima et al., 2012), which are the network flows with only one origin and one destination. Askoura et al. (2011) attempt to elicit a sub-network that is free of the BP using an optimization formulation, which seems adequate for single-commodity networks. However, for multi-commodity cases (travel demand involving many origin-destination pairs), a multi-criteria problem must be solved, which is yet to be investigated for large-sized transport networks.

The BPDP involves closing some roads while measuring the impact on the traffic circulation. This, in essence, makes the bilevel programming a natural structure to formulate the BPDP. Ma et al. (2017) formulate the BPDP as a bilevel problem, adopting a SO-relaxation approach for which a linear approximation method is developed. To account for the lower-level sub-problem, the traffic flow is expressed as an origin-based multi-commodity formulation, which becomes computationally cumbersome for large-sized networks. As a case study, the authors use the Sioux-Falls network (24 nodes). Yin and Ieda (2002) propose a bilevel programming model to avoid the reliability paradox for the existing roads, which is numerically tested using several toy networks. Bagloee et al. (2013a) propose a metaheuristic method (ant colony hybridized with the genetic algorithm) addressing the BPDP for a large example. The proposed method has two shortcomings: (i) It requires a pre-screening phase to flag some suspiciously Braess-tainted roads to be later vetted through a bilevel programming structure, and (ii) It ignores endogenous demand elasticity due to changes of the road network as a result of the roads closure. The above literature review underscores the following remarks:

- The BP is omnipresent in both theory and practice.
- Any changes to the network, from signal timing to adding new roads, and any changes in the travel demand might spur emergence or removal of the BP.
- The intensity of the BP is significantly correlated with the type of delay functions, and it can be extremely high.
- Network intervention schemes such as road closure are found effective to mitigate the BP.

- Detecting the BP in real-life examples is an arduous task (NP-hard problems).
- Studies to address the BPDP tailored to the large-sized network are scarce.
- Despite the importance of the travel demand in the BP, travel demand elasticity has been largely ignored in the previous studies.

To address these shortcomings, in this study, we address the BPDP for large-sized networks as a bilevel problem, while accounting for the demand elasticity. Given the authors' previous work (Bagloee et al., 2013a), the min contribution of the current study is twofold (i) inclusion of the demand elasticity, and (ii) proposing a much faster algorithm which has two practical features, namely being modularized, and being parallelized. The proposed algorithm is modularized in the sense that, the traffic simulation part can be replaced with any other meso or micro model (including a dynamic assignment) depending on the size and nature of the case-studies as well as the computational power. It is also a parallelized algorithm as the computations can be divided into several parallel channels in the interest of expediting the process. In the next section, we provide a bilevel formulation for the BPDP.

3. Mathematical formulation of the BPDP

Consider a road network and a list of the roads to be closed. Obviously, closing a road adversely affects accessibility to businesses and households that are directly served by the respective road, which needs to be considered as part of the cost of the road closure. In other words, one can interpret it as a business-loss cost imposed on local businesses and residents. Those who are affected by the road closure may then be compensated by schemes such as mobility credits, free public transport tickets, discounted parking permits in the central business district (CBD) etc. Hence, there must be a limited budget representing the authorities' capacity to afford these business-loss expenses. Nevertheless, this budget should also afford the costs involved in transforming the respective roads into a pedestrian plaza, green space, recreational facilities, and the like. Consequently, the BPDP involves identifying the subset of these listed roads (also known as candidates) to be closed with the aim of minimizing the total travel time for the system. Obviously, arriving at a commensurate assessment for the business-loss expenses could be a challenging task and is left for further studies

Table 1
Notations.

A, A'	Sets of existing roads and candidate roads, respectively
B	Budget to cover the business-loss expenses as well as the costs involved in transforming roads that are closed into a pedestrian plaza, parks, etc.
$B^{\min}$	The minimum budget that forces the algorithm to render more roads for closure
y_{ij}	The binary decision variables of the candidate road $(i, j) \in A'$; it takes the value of 1 for the respective candidate road decided to be closed, and 0 otherwise
c_{ij}	The business-loss cost involved in closing a candidate road $(i, j) \in A'$
x_{ij}	A continuous variable denoting traffic flow on the road $(i, j) \in A \cup A'$
$t_{ij}(x_{ij})$	Volume-delay function (VDF) or travel cost of the road $(i, j) \in A \cup A'$. We adopt the widely used BPR function (Bureau of Public Road (BPR, 1964)), which is $t_{ij}(x_{ij}) = t_{ij}^0(1 + .15(x_{ij}/v_{ij})^4)$. Note that, t_{ij}^0 is free-flow travel time (minutes) and v_{ij} is the capacity (vehicles per hour) of the respective road
W	Set of origin-destination (OD) pairs
q_w	Travel demand pertinent to origin-destination $w \in W$, pertaining to the elastic travel demand
P_w	Set of paths between origin-destination $w \in W$
h_p	Total flow on the path $p \in P_w$
$\delta_{ij,p}$	Link-path incident index, 1 if link $(i, j) \in A \cup A'$ belongs to the path p, and 0 otherwise
M	A sufficiently large value
Z	Total travel time (the objective function)
t_w	Minimum travel time (of the shortest path) for OD pair $w \in W$
$\tilde{Z}$	Linear regression function, as a surrogate function for the original objective function Z
$\bar{q}_w$, $\bar{t}_w$	Travel demand and travel time of the shortest path of the origin and destination (OD) $w \in W$ pertaining to the do-nothing scenario with fixed travel demand
$D(t_w)$	Elastic travel demand function for OD pair $w \in W$
$\Omega(q_w)$	The inverse of elastic travel demand function for OD pair $w \in W$, $\Omega(q_w) = D^{-1}(t_w)$
BP	Braess Paradox
k	Iteration number of the solution algorithm
α	The current iteration of the solution algorithm
$\alpha_{\max}$	Maximum number of iteration in the solution algorithm
Z^{α}	Value of the objective function (1) given a binary solution pertaining to iteration α
OD	Origin-destination
CBD	Central business district
BPDP	Braess Paradox Detection Problem
TAP	Traffic assignment problem
PA	Price of anarchy
BPDP-Fix	BPDP with fixed travel demand
BPDP-Els	BPDP with elastic travel demand
MILP	Mixed integer linear program
UE	User equilibrium
SO	System Optimal
ILP	Integer linear programming

(Dalziell and Nicholson, 2001). Let us first define the notation used in the formulation, as presented in Table 1.

We first, formulate the bilevel BPDP when the travel demand is a fixed matrix:

$$\text{BPDP} - \text{Fix: } \underset{x_{ij}, y_{ij}}{\text{Minimize}}\ Z(x_{ij}, y_{ij}) = \sum_{(i,j)\in A\cup A'} x_{ij}.\ t_{ij}(x_{ij}) \tag{1}$$

subject to

$$\sum_{(i,j)\in A'} c_{ij}.\ y_{ij} \leqslant B, \tag{2}$$

$$y_{ij} = \{0, 1\} \qquad (i, j) \in A' \tag{3}$$

$$\underset{x_{ij}}{Minimize} \sum_{(i,j)\in A\cup A'} \int_0^{x_{ij}} t_{ij}(x)\,dx \tag{4}$$

subject to

$$\sum_{p\in P_w} h_p = q_w, \qquad w \in W, \tag{5}$$

$$x_{ij} = \sum_{w\in W} \sum_{p\in P_w} h_p \delta_{ij,p}, \qquad (i, j) \in A \cup A', \tag{6}$$

$$x_{ij} \geqslant 0, \qquad (i, j) \in A \cup A'. \tag{7}$$

$$x_{ij} \leqslant M.\ (1 - y_{ij}) \qquad (i, j) \in A', \tag{8}$$

Eq. (1) minimizes the total travel time at the upper level. Constraints (2) and (3) guarantee feasibility of the binary solution ($\forall\ y_{ij}, (i, j) \in A'$) considering the costs and the budget. The lower level ((4)–(8)) is the traffic assignment problem (TAP), expressed based on Beckmann's user equilibrium (UE) formulation (Beckmann et al., 1956). Constraint (8) lets no traffic along a road that is decided to be closed (i.e., $y_{ij} = 1$).

$$M = \sum\nolimits_w q_w \tag{9}$$

In the above formulation, the travel demand is considered fixed, irrespective of any changes to the transport network and has no regards to the ensuing changes of travel times as the result of roads closures. Obviously, some commuters are highly sensitive to their travel times and have the flexibility to change their routine trips, such as changing their departure times or their usual mode of transport or even to forego their trip. Hence, to enhance fidelity and realism of the model, it is important to endogenously include variability of the travel demand in response to the effects of the road closures. The BPDP-Fix (BPDP with fixed travel time) formulation can be modified to include travel demand elasticity as follows:

$$\text{BPDP} - \text{Els: } \underset{x_{ij}, y_{ij}, q_w}{\text{Minimize}}\ Z(x_{ij}, y_{ij}) = \sum_{(i,j)\in A\cup A'} x_{ij}.\ t_{ij}(x_{ij}) \tag{10}$$

subject to

$$\sum_{(i,j)\in A'} c_{ij}.\ y_{ij} \leqslant B, \tag{repeated-2}$$

$$y_{ij} = \{0, 1\} \qquad (i, j) \in A' \tag{repeated-3}$$

$$\underset{x_{ij}, q_w}{Minimize} \sum_{(i,j)\in A\cup A'} \int_0^{x_{ij}} t_{ij}(x)\,dx \ - \sum_{w\in W} \int_0^{q_w} \Omega(\varphi).\ \varphi \tag{11}$$

subject to

$$\sum_{p\in P_w} h_p = q_w, \qquad w \in W, \tag{repeated-5}$$

$$x_{ij} = \sum_{w\in W} \sum_{p\in P_w} h_p \delta_{ij,p}, \qquad (i, j) \in A \cup A', \tag{repeated-6}$$

$$x_{ij} \geqslant 0, \qquad (i, j) \in A \cup A'. \tag{repeated-7}$$

$$x_{ij} \leqslant M.\ (1 - y_{ij}) \quad (i, j) \in A', \tag{repeated-8}$$

As can be seen, the travel demand q_w is no longer a fixed input, but rather, it is endogenously determined along with the traffic volumes x_{ij}, where both are dependent on the travel time t_{ij}. In contrast with (4), the objective function (11) has a new term which accounts for the elastic demand while the rest of the constraints remain unchanged.

Note that these bilevel formulations are basically the Stackelberg leader-follower games (Stackelberg, 1952). In fact, the road closure schemes and Braess Paradox can be studied in the field of game theory (Colombo and Holden, 2016). The leader (in this case, the transportation authority) first decides which roads must be closed (a solution for the binary variables y_{ij}, or, as we refer to them, decision variables). The followers (drivers) will then change their routes, seeking the new shortest paths considering the new changes vis-à-vis the network's accessibility, which results in roads' traffic flow x_{ij} to be computed by solving a TAP. Hence, given a feasible binary solution (y_{ij}), the BPDP rolls over to become simply a TAP. The TAP is a convex problem and can be efficiently solved using the off-the-shelf commercial software. Moreover, the fixed-demand TAP can be solved using a popular method known as convex combination also known as Frank–Wolfe algorithm, whereas the elastic–demand TAP is solved using the Frank–Wolfe algorithm and a partial linear approximation method (Florian, 1986). The above problem is convex for which there exist a variety of methods, such as Frank-Wolfe (or convex combination), (Patriksson, 2004), variational inequality (Dafermos, 1980, 1982), complementarity methods (Aashtiani and Magnanti, 1981; Bagloee et al., 2017c). Moreover, Sheffi (1985) has also discussed an interesting analytical interpretation of the elastic traffic assignment, that is, one can transform the elastic assignment to an equivalent expanded network to be solved simply as a normal traffic assignment. To solve both the fixed and elastic demand TAPs, we use EMME 4, one of the most computationally efficient software.

The crux of the above problems is to find a feasible solution for the binary decision variables for which we develop a surrogate-based technique, as discussed in the next section. For the sake of clarity and ease of presentation, we first provide an algorithm to solve the BPDP-Fix which is then generalized to solve the BPDP-Els (BPDP with elastic travel time).

4. Surrogate-based solution algorithm

The algorithm consists of a supervised learning technique as a proxy for the original objective function (function (1) in the BPDP-Fix) and an integer linear programming (ILP) problem to render a new solution for the binary decision variables.

The algorithm is ignited with the do-nothing scenario (the existing situation), a feasible binary solution ($y_{ij} = 0,\ \forall\ (i,j) \in A'$) for which the TAP is solved to compute a value of the objective function (1), denoted by Z. This set of data (Z and y_{ij}) is then used to train a multivariate linear regression model, which is a function of the decision variables:

$$\bar{Z} = \sum_{(i,j)\in A'} b_{ij}.\ y_{ij} \tag{12}$$

where $\bar{Z}$ is a linear approximation as a surrogate function of the original objective function Z, and b_{ij} are parameters to be calibrated based on the training data (Z and y_{ij}). The calibration can be easily done using statistical or econometric software (we use MATLAB).

The next step is to find a new and feasible solution for the binary decision variables via solving an ILP problem. Assuming that the surrogate algorithm postulates that the newly calibrated $\bar{Z}$ (function (12)) is a linear approximation of the original function (1) and it represents all the binary constraints of the original problem, the ILP of the BPDP-Fix can be written as:

BPD-Fix-ILP:

$$Minimize\ \bar{Z} = \sum_{(i,j)\in A'} b_{ij}.\ y_{ij} \tag{13}$$

subject to

$$\sum_{(i,j)\in A'} c_{ij}.\ y_{ij} \leqslant B, \tag{repeated-2}$$

$$y_{ij} = \{0,\ 1\} \qquad (i,j) \in A' \tag{repeated-3}$$

$$\begin{cases} \sum_{(i,j)\in Y1^k} y_{ij} - \sum_{(i,j)\in Y0^k} y_{ij} \leqslant\ |Y1^k| - 1 \\ where: \\ Y1^k = \{(i,j)|y_{ij}^k = 1\};\quad Y0^k = \{(i,j)|y_{ij}^k = 0\} \quad (i,j) \in A',\ \ k = 1 \ldots \alpha \end{cases} \tag{14}$$

Constraint (14) ensures that a binary solution out of the BPDP-Fix-ILP for the next iteration $\alpha + 1$ is new $y_{ij}^{\alpha+1}$, and it has not been generated in the previous iterations (Balas and Jeroslow, 1972). Note that k denotes the iteration number, α is the current iteration, and $Y1^k,\ Y0^k$ are the sets of binary variables that have taken values of 1 and 0 pertaining to iteration k, respectively.

For the next step, the new binary solution ($y_{ij}^{\alpha+1}$) is implanted into the original BPDP-Fix (to turn it to a TAP). When the TAP is solved, the total travel time ($Z^{\alpha+1}$) is also computed, which provides a new training record of ($Z^{\alpha+1}$, $y_{ij}^{\alpha+1}$) for the next regression. Hence, for each iteration, the algorithm calibrates a new regression function (12), followed by solving the BPDP-Fix-ILP and the TAP. This process continues until $\alpha_{\max}$, a pre-specified maximum number of iterations, is exhausted. Empirical evidence suggests setting $\alpha_{\max}$ to no less than 5 times the total number of binary variables (Bishop, 2006).

As noted before, the BPDP-Fix-ILP is, in fact, an approximation of the BPDP-Fix, in which the regression function (13) represents the original objective function (1) and the constraints related to x_{ij}, the drivers' route choices. In other words, the traffic flows have been dropped in the BPDP-Fix-ILP, resulting in significantly fewer variables and constraints, which in turn improves the computational efficiencies. Note that, all the variables of the BPDP-Fix-ILP are binary, and all the functions (i.e., the constraints and the objective function) are linear. This would expedite the algorithm's runtime and makes it amenable to commercial optimization

software to be solved efficiently as an ILP problem.

In the end, the binary solution with the minimum value of the objective function is reported as the best solution found. Let us recap this process as a formal algorithm.

- Step 0, initialization: Set iteration counter $\alpha = 0$, the maximum number of iterations $\alpha_{\max}$, and the do-nothing scenario as an initial feasible solution $y_{ij}^{0} = 0, \quad (i, j) \in A'$.
- Step 1, traffic assignment: With respect to the BPDP-Fix, given the current feasible solution $y_{ij}^{\alpha} = 0, \quad (i, j) \in A'$, solve the resulting TAP and compute the value of the objective function Z^{α}.
- Step 2, surrogate function: Given $Z^{k}, \ y_{ij}^{k}, k = 1..\alpha$, calibrate $\bar{Z}^{\alpha}$ a new regression function (12).
- Step 3, update/run integer problem: Update the objective function (13) and constraint (14) in the BPDP-Fix-ILP and solve it to find the next and new feasible solution $y_{ij}^{\alpha+1} = 0, \quad (i, j) \in A'$.
- Step 4, termination criterion: If $\alpha < \alpha_{\max}$, then set $\alpha = \alpha + 1$ and go to Step 1; otherwise, stop and report (i.e., $y_{ij}^{*}, \ Z^{*}$ where $Z^{*} = \text{argmin}_{\ k=1..\alpha_{\max}} Z^{k}$). ■

Additionally, the surrogate algorithm is a parsimonious and frugal algorithm that imposes no specific requirements on the original problem. In fact, it treats the original problem as a black box, that is, the only important things are the input (i.e., the decision variables y_{ij}) and the output (i.e., the value of the objective function Z) based on which the algorithm builds a regression model to be used in the ILP. Therefore, in the above exposition, the BPDP-Fix can simply be replaced with the BPDP-Els.

5. Numerical tests

In this section, we discuss the numerical results of the proposed methodology applied to the large-sized network of Winnipeg, which consists of 154 traffic analysis zones, 944 nodes, and 3383 directional links. The fixed travel demand is a 154-by-154 matrix of 56,804 hourly trips. We have nominated all the roads belonging to the central business district (CBD) as candidates for closure, amounting to 801 roads spanning a total of 430.64 km as shown in Fig. 2. With no loss of generality, for the ease of presentation, we consider the length of the roads, which vary from 0.06 km (or 60 m) to 2.7 km with an average of 0.54 km, as their respective business-loss cost, if they are closed. We have made this dataset available to the research community via a GitHub link to be used as a benchmark case in future studies (Bagloee, 2018).

The TAP is the building block of the analysis, which needs to be computed with a very fine precision rate. To this end, as suggested by Boyce et al. (2004), we set the "relative gap" of the TAP to be 0.0001, which is a fine convergence rate between successive iterations of the Frank–Wolfe algorithm to be solved by EMME 4. Hence, the algorithm is coded in synchronization with the EMME 4 and is linked to MS-Excel as an interface as well as MS-Access database. The code utilizes GAMS (Baron solver (Tawarmalani and Sahinidis, 2005)) to solve the ILP along with MATLAB (MathWorks, 2016) to solve the multivariate regression. We use a desktop computer with 64.0 GB RAM and CPU processor of Intel Xeon 3.70 GHz.

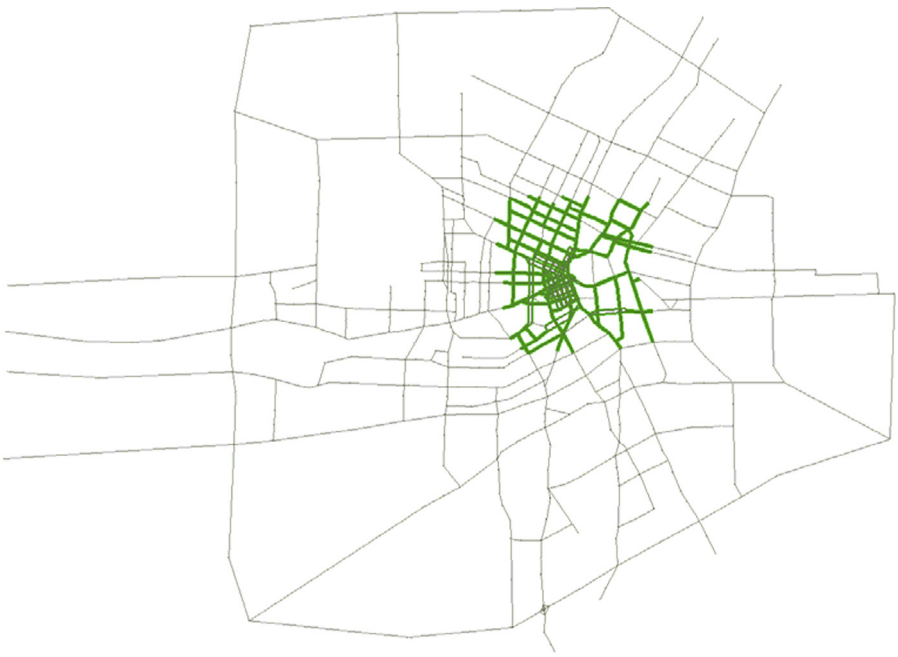

Fig. 2. Winnipeg – the extent of the road network and 801 roads that are candidates for closure.

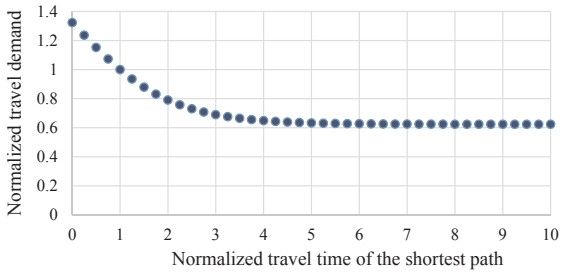

Fig. 3. Elastic travel demand function (travel time and travel demand are normalized in the sense that they are divided by their equivalent values of the no-closure scenario to render dimension-less values).

We first solved the TAP for the do-nothing scenario (no road closure) with the travel demand considered fixed, based on which the total travel time was found to be 827075.87. As for the elastic demand, we considered a logistic function as follows:

$$D(t_w) = q_w = \bar{q}_w \,.\, (0.62348 + 1.4 \frac{e^{-t_w/\bar{t}_w}}{1 + e^{-t_w/\bar{t}_w}}) \tag{15}$$

where $\bar{q}_w$, $\bar{t}_w$ are travel demand and travel time of the shortest path of the origin and destination (OD) $w \in W$ pertaining to the do-nothing scenario with fixed travel demand. Fig. 3 exhibits the function based on the normalized values of the travel demand (i.e., $q_w/\bar{q}_w$) versus the normalized travel time (i.e., $t_w/\bar{t}_w$) of the shortest path between the respective OD. The elastic demand function (15) is calibrated in such a way that, it yields the very same fixed travel demand when the travel times are identical to those of the do-nothing scenario (i.e., $q_w = \bar{q}_w$, if $t_w = \bar{t}_w$, $\forall\, w \in W$). Moreover, Eq. (15) complies with the general format of the logit models widely used in travel mode choice practices.

Moreover, in the worse- and best-case scenarios, the elastic travel demand would vary between 0.62 and 1.32 of the fixed travel demand, respectively. In other words, if all roads are closed the travel demand does not equate to zero. In actual fact, the minimum travel demand is equal to 62% of the fixed travel demand, the do-nothing scenario.

To start the numerical experiments, we first solve the BPDP for the fixed travel time (BPDP-Fix) while omitting the budget constraint. This is to determine whether the BP exists in the Winnipeg case study. Given the number of binary decision variables (801), as noted before, the maximum number of iterations $\alpha_{\max}$ is set to be $4005 = 5 * 801$. Fig. 4 provides a snapshot of the algorithm in terms of the way the value of the objective function converges over successive iterations. As can be seen from the green-colored dots, the fluctuation of the outcome in early iterations, when the regression's training is underway, is high. However, as the algorithm proceeds to more iterations, these drastic variations dampen, and they stabilize in final iterations. The red-colored graph in Fig. 4 shows the progressive improvement in the value of the objective functions (or the best solutions found). The best (minimum) value of the objective function is found to be 826693.06, which is the result of closing 10 roads with a total length of 3.8 km (see Fig. 5). It is worth noting that closure of these 10 roads adds no additional burden on the existing traffic congestion; in fact, it improves it (i.e., the total travel time becomes lower, $826693.06 < 827075.87$).

We can now run the algorithm for an elastic travel demand scenario (BPDP-Els). Similarly, we first solve the BPDP-Els with no budget constraint, based on which the algorithm identifies 40 road segments, of a total length of 10.42 km (see Fig. 6). The total travel time of this scenario was found to be 826878.93, slightly higher than that of the fixed demand scenario (i.e., $826878.93 > 826693.06$), which is due to the fact that by closing some Braess-tainted roads, the travel time for some OD pairs decreases, which induces additional travel demand, as suggested by the elastic demand function (15).

Since the main motive of this study is to boost the notion of reclaiming road space, for the next experiment, we force the algorithm to render as many roads as possible for closure, without degenerating traffic congestion. To this end, we replace the budget constraint (2) of the BPDP-Els with the following constraint:

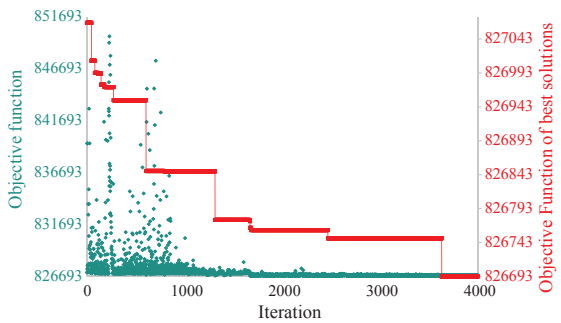

Fig. 4. Fixed-demand scenario – variations of the objective function over successive iterations.

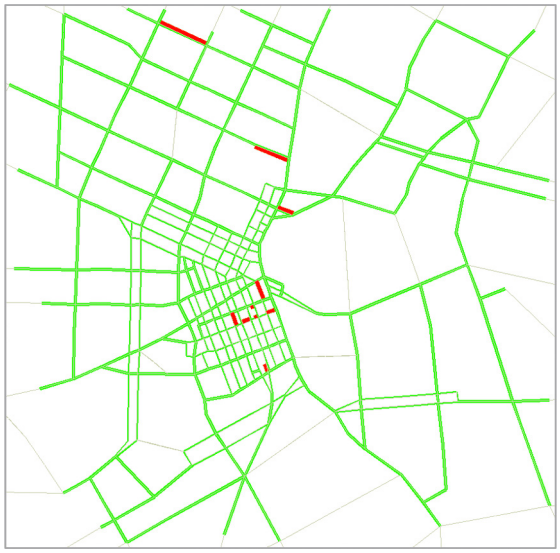

Fig. 5. Fixed-demand scenario – 10 Braess-tainted roads to be closed.

Fig. 6. Elastic demand, with no budget constraint – roads to be closed.

$$\sum_{(i,j)\in A'} c_{ij} \cdot y_{ij} \geqslant B^{\min}, \tag{16}$$

where the algorithm has to generate scenarios whose total costs are equal to or greater than a minimum budget denoted by $B^{\min}$. The minimum budget needs to be assigned with a commensurate value such that the final outcome of closing some roads does not compromise the existing level of traffic congestion. To this end, let us compare the total travel times of the do-nothing (or existing) scenario with that of the elastic demand scenario with no budget, which are 827075.87 and 826878.93, respectively. It is evident that the total travel time after closing 10.42 km of roads is still less than that of the existing traffic congestion (i.e., 826878.93 $<$ 827075.87); hence, the minimum budget can be a value greater than 10.42 such that the total travel time of the final outcome would not be significantly greater than that of the existing scenario. Therefore, we set the minimum budget to be $B^{\min} = 12 > 10.42$. With these settings, we run the algorithm, which results in a scenario of the total travel time of 827615.93, corresponding to the closure of 20 roads with the total length of 12.16 km (see Fig. 7).

We conclude this section by discussing the calibration details of the algorithm, the computational burdens and a comparative analysis with a metaheuristic algorithm. Fig. 8 depicts variations of the objective functions of two elastic demand scenarios: with and without the budget constraint (16). As can be seen, both show a similar pattern, that is, the changes are high in early iterations and decrease (and converge) in final iterations. Based on the same dataset, Fig. 9 demonstrates the pace of improvements in the objective functions.

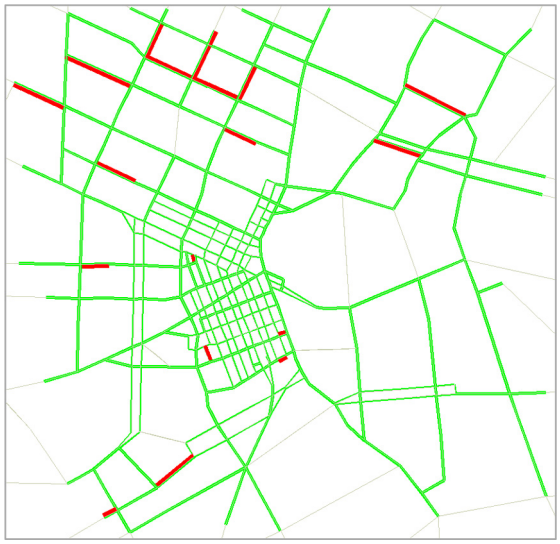

Fig. 7. Elastic demand, with a minimum budget of 12 – roads to be closed.

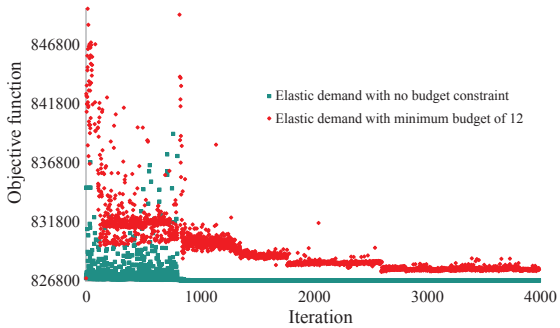

Fig. 8. Elastic demand, with and without budget constraint – variations of the objective function.

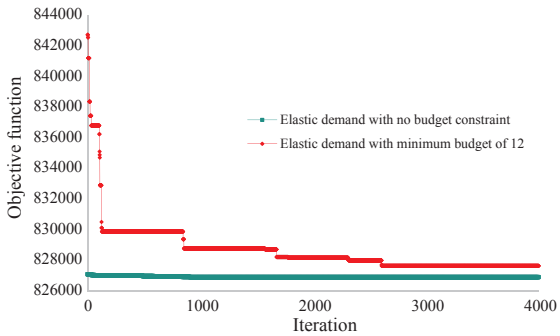

Fig. 9. Elastic demand, with/without budget constraint, gradual improvement of the objective function.

It is also important to investigate the calibration process in terms of the variation of the parameters over successive iterations. Fig. 10 provides an overview of these results by selecting 8 (out of 801) parameters corresponding to candidate roads 1 to 8 (note that, to have a clear snapshot, the figure reports on the normalized values of the parameters projected to an interval of 0 to 100). The figure suppresses the iterations that are less than 801, since the variations of the parameters are too high. According to Fig. 10, the algorithm exhibits a vivid convergence behavior, which can serve as confirmation.

As can be seen from Fig. 10, the regression model has reached a maturity level (convergence) at iteration 1800th onward which raises a valid question: why not to stop the training when the regression's parameters converge? The answer to this question can be subjective. Generally speaking, in comparison to solving the traffic assignment and solving the integer linear programs, training the regression part is not computationally hefty, hence, it is desirable to keep the training of the regression on. Otherwise, one can

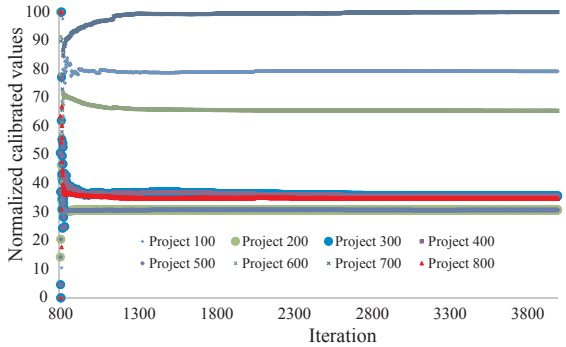

Fig. 10. Elastic demand with a minimum budget of 12 – calibration results.

expedit the computation by freezing the regression's parameters.

Furthermore, Fig. 11 shows the difference between the objective functions estimated by the regression function and the actual values computed after solving corresponding TAPs as an error index. As evidenced in the figure, this error can be as high as 1.4% in early iterations, but it decreases sharply in final iterations. We have also computed the root-mean-square error (RMSE) as an error index for the overall performance of the regression model, which was calculated to be 918.04. Considering the average of the real values of the objective functions (i.e., 827057), the RMSE is equivalent to 0.11% error (0.11% = 918.04/827057 * 100), which can be considered as a negligible error.

Figs. 8 and 11 exhibit a U-Shape pattern in early iterations before the algorithm converges (at the outset, these values improve slightly, but then they deteriorate and increase till a certain iteration, and they then promptly drop down again and stay at the convergence level). To explain this behavior, it is important to note that the problem itself is a minimization problem and the objective value Z is a (large) positive value. The algorithm starts with a zero-out objective function. In early iterations, those (binary decision) variables who are assigned value 1 (when the ILP is solved in Step 3), will then be given some large positive coefficients in the regression process (Step 2). Hence in the next iteration, when the minimization problem is solved (the ILP in Step 3), the binary variables with large positive coefficients in the objective function will not be selected and instead the binary variables with the coefficient of zero will be selected (and will be given the value of 1). Therefore, the algorithm is forced to first exhaust all the variables with zero coefficients in Z which results in a significant deterioration of the objective value until a certain iteration. This process depicts a U-shape. This is, in fact, a healthy setup to get the algorithm to explore the entire solution space. This U-shape part can be regarded as a training stage. When the training is consolidated, the algorithm generates some good performing solutions around optimal solutions (though we cannot certainly say that they are global optimal solutions). This is a very significant

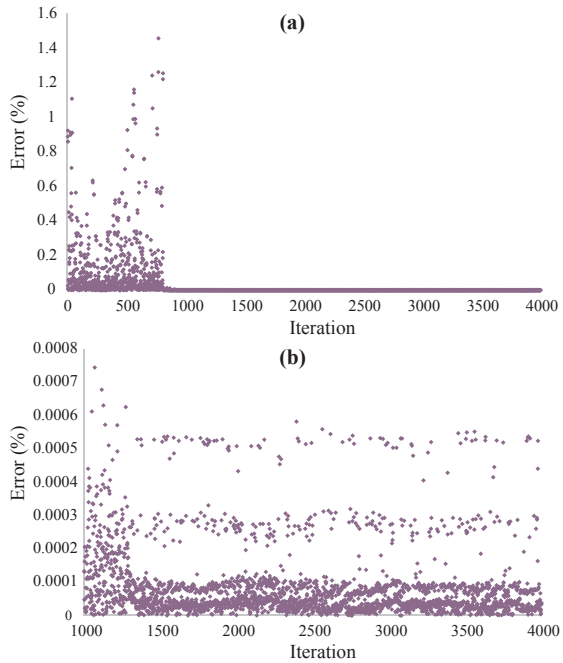

Fig. 11. Elastic demand, with a minimum budget of 12 – error index, (a) across all the iterations, (b) zoned in for iteration 1000 and above.

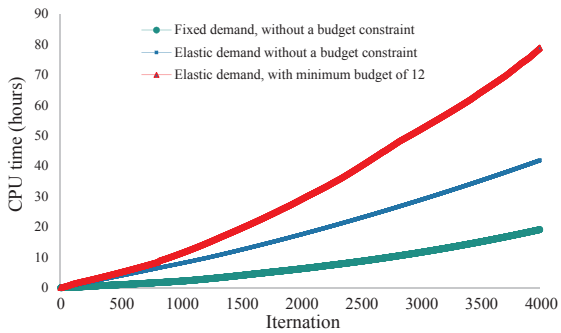

Fig. 12. Computational times of three scenarios.

observation that can be utilized to specify a convincing clue about the maximum number of iterations. As the result indicates, a modeler must first let the algorithm finishes the training stage and carry on for some additional iterations (e.g. twice the length of the training stage), before deciding to terminate the iterative process.

Fig. 12 displays CPU time of three scenarios: (i) fixed demand with no budget, (ii) elastic demand with no budget, and (iii) elastic demand with a minimum budget of 12, for which the total computational times are 19, 41 and 79 h respectively. Although, these CPU times may seem too high and unaffordable, with respect to the highly strategic nature of the problem, going for an even higher computational time is truly justified. In fact, nowadays in some engineering practices, an over-weekend computational time has become a unit of computation. Additionally, solving the BPDP is a one-off job, meaning that, it does not need to be solved frequently on a daily basis; hence a prolonged computation is not a deterrence factor for strategic problems. Moreover, it is important to underline the (almost) linear and non-exponential pace of the CPU time over incremental iterations in all three scenarios, which can serve as confirmation when facing the large-sized network to be solved in more iterations.

It is obvious that the inclusion of demand elasticity adds significant burden on the CPU time. This is due to the fact that solving for traffic assignment problem with variable demand (TAP-VD) is computationally expensive than that of the TAP with fixed demand (TAP-FIX). To have palpable glimpse, as noted before, a TAP-VD can be reformulated as an extended network to be solve simply as a TAP-FIX. To do so, Sheffi (1985) has shown that it suffice to add one additional node and one additional link (dummy node and link) for every origin-destination pair. Therefore, for the Winnipeg case-study which has 154 zones, 944 nodes, and 3383 links, the total number of nodes and links of the extended network (for TAP-VD) will be 24,660 and 27,099, respectively (e.g. $24660 = 944 + 154 \times 154$). As can be seen, this inflates the size of the problem (number of links) 8 times (i.e. $8.01 = 27099–3383$) which is very significant.

We conclude this section by reporting results of a comparative analysis with a hybrid metaheuristic method called Gen-Ant. The Gen-Ant is a combination of Genetic Algorithm and Ant Colony which has been successfully applied to a similar problem, namely the discrete network design problem (Bagloee et al., 2013b). We run the Gen-Ant for the scenario of elastic demand without a budget constraint. To set up a fairground for comparison, we run the Gen-Ant for 41 h, which is identical to the CPU time of the surrogate method. This, for the Gen-Ant, would result in solving the TAP for 12,000 times. Given the randomness nature of the Gen-Ant, we repeated it for 5 trial and report the average of the results. The average of the total travel time was found to be 827001.09 which is not better than that of the surrogate method (i.e. $827001.09 \not< 826878.93$).

6. Conclusion

Reclaiming road space has recently become a demand touted by environmental activists, especially in some European cities in which authorities have pushed for banning cars in some parts of cities. This has also coincided with a paradigm shift stemming from a tech-savvy and driving-shy generation, which has precipitated smart mobility and fewer cars on the roads, especially in some developed countries. Though the main motive of no-car schemes is largely rooted in environmental concerns, in this study, we tried to show that they can also help to alleviate chronic traffic congestion. In other words, closing roads to vehicles can sometimes improve traffic circulation, a phenomenon known as Braess Paradox. The idea is to find a number of Braess-tainted roads to be closed with the aim of alleviating traffic congestion, or, as it is called, the Braess Paradox Detection Problem (BPDP).

We formulated the BPDP as a bilevel problem to select a number of roads to be closed such that at the upper level, the total travel time (as an index for traffic congestion) is minimized while accounting for the changes in the traffic circulation as a traffic assignment problem at the lower level. The BPDP is a complex and difficult problem (i.e., NP-hard), and it becomes computationally prohibitive in real-life examples.

To cope with such a complexity and solve the BPDP for a real-life example, we developed a surrogate-based methodology which proves to be computationally efficient for large-sized networks. Obviously, drivers react to the changes in their travel time due to the impact of a road closure scenario by changing their routing departure time or travel mode or even canceling their trips. These changes affect the travel demand, which must be taken into account when solving the BPDP and assessing the impact of a road closure scenario. To this end, we solve for an elastic traffic assignment problem (at the lower level of the bilevel problem); that is, the travel demand is endogenously set as a function of the changes of the travel time. We solved the BPDP for a real dataset, the network of

Winnipeg based on which the algorithm managed to identify a number of roads to be closed under different operational scenarios without exacerbating the existing level of traffic congestion. In other words, these road closures do not add to the existing level of traffic congestion; in fact, they improve it, as the notion of the Braess Paradox shows. We showed that the proposed methodology can accommodate authorities' operational concerns, such as identifying more roads for closure.

Given the ever-increasing interest in car-banning policies, there remain several directions for further research: (i) It makes sense to take public transport as an auxiliary mode into account when accounting for the variation of the travel demand; (ii) Considering the stochasticity of BPDP inputs and parameters, such as costs, budgets, and travel time, can offer a more reliable outcome; (iii) From a mathematical point of view, partial closure is supposed to render more benefit than a full closure. To this end, the proposed methodology can be extended to lane-based closure, in the sense that for each lane of a directional link is associated with a decision binary variable. Hence, the problem transforms to whether to close a lane. Alternatively, the partial closure can be formulated also as a continuous problem, in the sense that, the decision variable would be linked to the capacity of the respective roads (e.g. for a road with the capacity of 1000 vehicle per hour, what would be the operational capacity, which is obviously anything below 1000?). This would invoke a new problem, called capacitated traffic assignment problem (Bagloee and Sarvi, 2015); (iv) Though the BPDP itself is a computationally difficult problem, considering a dynamic traffic assignment—that is, to investigate the traffic circulation in a much finer and detailed way—would further enhance the realism of the outcome; (v) Road closure can be coupled with effective congestion pricing; (vi) It is also a worthwhile effort to expand the methodology to multiclass traffic flow. For multiclass traffic flow, the logit function (15) can be integrated into the traffic assignment problem as a combined mode-choice and assignment model, which can then be solved with methods such as variational inequality and complementarity methods (Boyce and Bar-Gera, 2004). The downside with such exact methods is the excessive computational burden, hence, an alternative is to resort to heuristic methods such as the Method of Successive. Averages (Boyce and Florian, 2005).

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.tra.2019.01.021.

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